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Flow, Riverscape, & Come Home: Three Installation Designs for the PLaY+SPaCE Ultrasonic Sensing Environment

Abstract

Sound installation design presents numerous compositional challenges, not least of these the requirement that an installation is to be 'played' by the public. In presenting three sound installations undertaken with an ultrasonic sensing system, possibilities for sound installation design using non-tactile sensing systems within the public space are demonstrated. Each of the three installations has required a different design in sensor placement, audio assignment and spatialisation, the various designs indicative of the functionality of non-tactile ultrasonic systems in installation environments.

Installation Design

A diversity of approaches to sound installation design, particularly in Australia, is well documented in Bandt (2005, 2006). Bandt presents representative examples of installations that "include permanent public and ephemeral sculptures, time-dense computerised sound installations, museum designs, exhibits in airports, art galleries, car parks, digital and interactive media exhibitions, and real-time virtual habitats on and off the web" (Bandt 2005).

The three installations discussed here (*Flow*, *Riverscape* and *Come Home*) are all indoor temporary works, the first developed as a stand-alone sound installation within a festival, the second and third accompanying and augmenting community-based visual art works in exhibition settings. In their design, the three installations necessarily vary as a result of the size and acoustics of the installation spaces, and the nature of the audio outputs required in each setting (music, texts, environmental sounds etc). From one installation design to the next, these factors alter a wide range of design parameters including the number and placement of sensors within the installation space, the number of divisions or triggering points within a sensor's range, the number of speakers to be used, their configuration (mono, stereo or multi-channel) and placement in the space, the density of audio textures, selection of pitch and rhythmic materials, degrees of predictability and change, and levels of interactivity. In the following overviews, these parameters are discussed in relation to the design of each installation and the installation setting.

Sensing Environment

The three installation designs discussed are based on the use of the PLaY+SPaCE ultrasonic sensing environment. The PLaY+SPaCE system uses up to eight ultrasonic sensors to detect positions of people within each sensor's detection range, each sensor individually detecting human positions over a range of 10 meters, with a minimum resolution of 15cm per position. The sensors use a proprietary hardware interface to map sensor data to MIDI. A software interface developed in *MAX/MSP* is then used to map incoming MIDI data. The mapped data is subsequently used in further *MAX* patches to control audio/music outputs designed for each individual installation. Further technical attributes and the design of the overall system have been detailed previously in Campbell (2003, 2005).

As a non-tactile system, the PLaY+SPaCE environment allows installation visitors/ participants to move within the space unencumbered by any physically attached sensing devices or wiring. The eight sensors of the system are generally located to form a 4 x 4 grid within a 100sqm sensing space, the sensors placed at (adult) waist height to detect movements of people moving to and away from each sensor across the space. This grid configuration was used in the first installation (*Flow*). Sensor location is not however limited to grid placements as each sensor may be placed in any location within an installation space, this allowing a considerable level of flexibility in installation design. The second installation discussed (*Riverscape*) used such an alternative approach to present three separate audio environments within a single installation space. The third installation (*Come Home*) used only two sensors placed on a single wall to create a sensing space within a small room.

Audio output configurations necessarily alter from one installation to the next, the three installations discussed here using a stereo system (*Flow*), a quad system (*Come Home*) and a stereo combined with two mono output configurations (*Riverscape*).

Whilst the discussion here is focused on the PLaY+SPaCE system, it is possible to apply the presented installation designs to alternative non-tactile systems and achieve similar results. Such systems could be based on, for example, pressure sensitive mats or video-based sensing.

Reactive Sound

From an artistic and compositional perspective, sound installations provide challenging pursuits for the audio artist in that they must be 'played' by the general public. The three installations discussed here are all 'interactive' in the most elemental sense of the term, though as discourse regarding interactivity in new media and musical contexts suggests (Bongers 1999, Manovich 2001, Winkler 1998), each of the three installations discussed relies on 'reactivity' as opposed to interactivity.

In essence, 'reactive' systems do not rely or expand on collected or predicted data (e.g. past or possible future states) to influence the current state of the system. More simply, reactive systems singularly output data in response to single data inputs. Such systems do however provide sufficiently entertaining, engaging and rewarding outputs for the public, particularly when the public is unfamiliar with the technology and the manner in which to engage with it. In many sound installations presented for the use of the public, a simple reactive design is indeed a necessity, required in order to actively engage the public with the themes and contexts of an installation.

Installation 1 – 'Flow'

Flow was designed as a work for inclusion in an annual regional festival (Riverfest) and was the inaugural public presentation of the PLaY+SPaCE system in 2002. The installation was presented in a bare space (c.80sqm) on the festival site, as shown in Figure 1. This was a purely audio installation, and with the exception of written directions for the public to explore the sounds presented through their movement through the space, no further media was provided.



Figure 1 – 'Flow' Installation Space

The work utilised a mode developed for the system in which the audio output is based on the measurement of activity levels within the space over two second periods (called 'Activity Mode'). Ten activity levels are recognized in this work, each relating to a musical environment based on differing tonal harmonic structures, rhythms and textural densities.

The work has a programmatic basis of a river and water flow. It has a very simple premise in that as activity levels increase in the physical space, so too do the levels of musical activity, these associated with the increase in the flow of a river from a calm and quiet start through to white-water rapids. The program uses various samples of water, marimba, vibraphone, percussion and a plucked piano bass, the timbral combinations, dynamic levels and harmonic tension increasing with the levels of physical activity, as shown in Figure 2.

Activity	Texture	Dynamic	Harmony
1	**V**ibes/**W**ater	pp	I
2	**P**erc/W	pp	n/a
3	V/P/W	p	Ii
4	**M**arimba/P	p	IV
5	All	mf	V
6	All	mf	V
7	All	f	I
8	All	f	Vi
9	All	ff	Ii
10	All	ff	I

Figure 2 – Flow Activity Levels and Musical Settings

In keeping with the programmatic basis of an ever-changing river, the differing timbral and rhythmic textures are generated in the Flow patch using a variety of controls to present sample sets at each Activity Level that appear to be constantly changing. In the case of rhythm, different combinations of (quasi) random numbers and altered metronome triggering speeds are applied at each activity level to generate rhythmic variations.

As an example, the Percussion sample set at Activity Level 2 is comprised of two conga samples, assigned with index numbers 1 and 2. A random generator for the samples provides numbers from 1 to 10 at this Activity Level, so that only the numbers 1 and 2 provide audible percussion outputs while the numbers 3 through 10 provide silence. With such a configuration the texture tends to be sparse. At the high Activity Levels of 9 and 10 the sample set is comprised of nine percussion samples. Random number generation is again set to provide numbers from 1 to 10, resulting in only one of the ten random numbers providing silence.

Additionally, at the higher Activity Levels, random numbers are used to provide changes in note triggering speeds. At the lower Activity Levels the periodicity of random number generation is set at 250ms. An underlying and ever-present ostinato in the work is set at 1000ms, set-

ting a tempo of ¼ = 60bpm. The note length at the lower Activity Levels is thus 1/16th. At the higher Activity Levels the note length is randomly changed from 250ms to 125ms (1/16th to 1/32nd notes) providing constantly changing patterns in both rhythm and timbre.

The linear programmatic basis of this work, moving from calm waters to rapids, provides a musical architecture that will inevitably take an arch shaped form as participants strive to attain the 'rapids' then relax their movement. With the relaxation of the participants after reaching 'the rapids', the system responds with less dense textures and rhythms, and a tonal harmonic setting that eventually returns to tonic harmony.

Whilst 'Flow' was originally designed for a river festival, the design of the work results in it being applicable in further installation settings, so long as the visitor is able to relate the sounds and activity of the audio output to the river-based programme.

Installation 2 – 'Riverscape'

Riverscape was part of a 2003 exhibition commissioned by the Townsville City Council, with support from the Australia Council and Delphin. The project was multi-disciplinary, with professional artists working in conjunction with amateur artists resident in the Riverside Gardens suburb of Townsville.

The culmination of the project was the exhibition of various art works developed by nine local participants working with professional artists. A series of poems, photographic works, nine 'Concept Maps' (mixed media visual art works), banners (displaying original poetry by the participants), a 'Ribbon Map' (an aerial photo adorned with participants' names and ribbons illustrating their daily movements in and out of the locality) and soundscapes (derived from interviews with the participants and recordings made within the local environment) were all included.

The relatively small area of the exhibition space, the live acoustics of the room (polished wood floors and numerous windows) and the quantity of exhibition materials limited the number of individual audio areas available in the space, and the size of each area. The resulting assignment of audio materials was to three audio areas, relative to one of the poem banners, two of the 'Concept Maps', and the 'Ribbon Map' as shown in Figure 3.

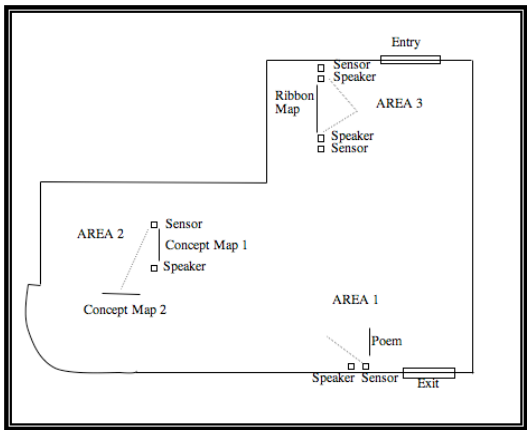


Figure 3 – Riverscape Sound Areas

In Area 1 (the poem banner) the sensor's range is divided into two triggering points. As a viewer approaches the banner, a range of environmental sounds is triggered, relative to the content of the poem. Moving closer to the poem, the viewer activates a second triggering point resulting in the playing of a recorded reading of the poem by its author. Should the viewer remain stationary, the poem will be heard in its entirety. If the viewer moves out of the sensing area for less than five seconds the reading will pause at the end of a stanza and resume on reentry at the beginning of the next stanza. Inactivity lasting any longer than five seconds resets the recording to the start. The purpose of this was to allow visitors a short time to reflect on each stanza of the poem.

In Area 2 the 'Concept Maps' are accompanied by written texts by the artists explaining the artworks. Concept Map 1 has a central theme of transition from a life/career on the sea to life on land. Concept Map 2 is by a Kenyan immigrant and relates the journey to Australia via China.

The proximity of the two artworks to one another allowed the use of a single sensor in the area, the range of the sensor divided into six triggering points as shown in Figure 4. The trigger points nearest the artworks (Figure 4 points 2 and 3 for Map 1, and points 5 and 6 for Map 2) were assigned for audio output.

Audio outputs relative to Concept Map 1 included thirteen samples from a recorded interview with the artist, along with water-based environmental sounds relative to the artist's theme. Trigger points 2 and 3 are assigned to trigger both the interview and environmental sounds. Similarly, Concept Map 2 sounds are based on interview excerpts and further water-based environmental sounds to represent the artist's ocean journeys.

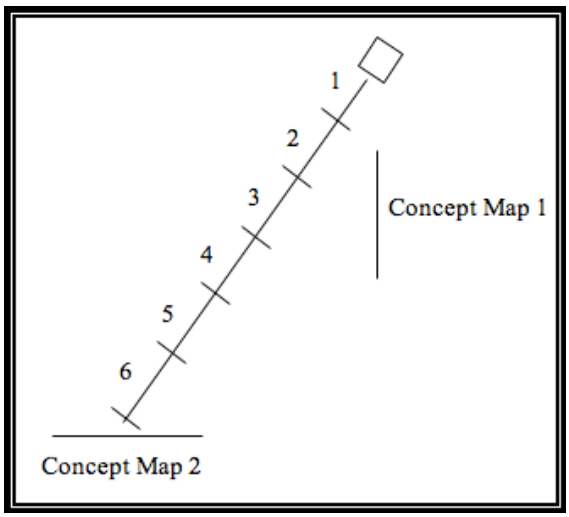


Figure 4 – Riverscape Area 2, Sensor Triggering Points

Figure 5 shows the 'Ribbon Map' placed in Area 3 of the installation. The large size of the map, in comparison to the artworks in the other areas, facilitated the use of two sensors, these placed so that the sensor beams crossed, as shown in Figure 6.



Figure 5 – Riverscape Area 3, Ribbon Map

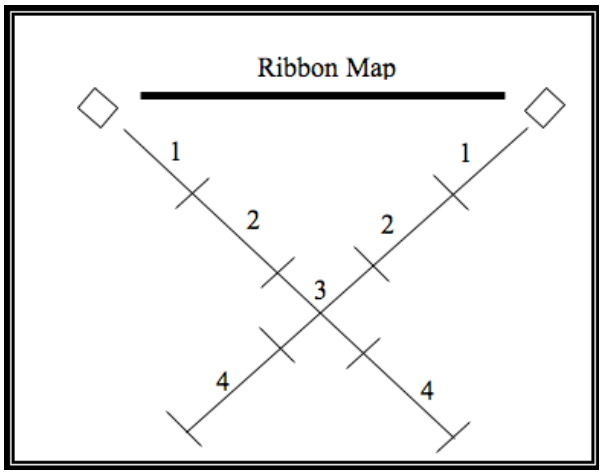


Figure 6 – Riverscape Area 3, Sensor Triggering Points

Due to the proximity of the sensors to the peripheries of the map, point 1 of the range is not used to trigger samples. Point 2 of the range, of both left and right sensors, is assigned to a series of 31 brief samples of the nine participants' voices. The text content is primarily focussed on life in the participants' locality, and relative to the content of the Ribbon Map.

Part 3 of the sensor range is assigned to children's voices either reciting or singing, in alphabetical order, the names of streets in the locality. Due to the central placement of the part in the Ribbon Map area, these samples were assigned a probability of 25% in occurrence, i.e. one in four triggers would result in the samples being heard. Additionally, the samples were faded in and out over a five second period in order to avoid abrupt intrusions to the text samples.

Part 4 of the sensor range triggers environmental or ambient sounds, of which there are 44 individual samples. Content of these sound files included sounds of bicycles, commentators on boat rowing competitions, sounds of tennis matches, and so forth. These samples were triggered as visitors entered through the front door, along with the occasional children's voice samples. Moving closer towards the Ribbon Map resulted in the triggering of the participants' voice samples.

The samples triggered by both parts 2 and 4 were also panned in the stereo field. When a viewer triggered a sample with the left sensor, the sample would be heard from the left speaker, and similarly from the right speaker when the right sensor was triggered. As most of the samples were a number of seconds in length, the sample would be heard in its entirety before another trigger could be heard. However, if the viewer, having triggered a sample on one side and hearing it through the corresponding speaker, then moved to the other side of the map, the currently playing sample would pan to the opposite speaker to 'follow' the movement of the viewer.

Due to the proximity of the three areas within the exhibition and the live acoustic of the space, problems were encountered with audio spill from one area to another. These were overcome however through speakers being placed/angled in such a way as to minimise interference from one area to the next. Whilst audio from one area could certainly be heard in another, the clarity of the sound samples and the placement of the speakers provided a clear audio experience relative to each individual area, the audio from the other areas generally being indistinct and heard as background sound.

The incorporation of sound into the exhibition heightened the experience for the visitor by providing audio that the visitor could control. Further, the recordings of the artists' interviews and the reading of the poem allowed visitors to make a greater connection with the artists and their work.

Installation 3 'Come Home'

Come Home was a collaborative work (2007) involving a group of professional artists working with children from three Townsville primary schools to develop artworks for an exhibition in a small empty house. The children were asked to identify an elder and under the guidance of the artists, to develop artworks based on their chosen elder and their relationship with that elder.

Stories from the children were recorded with the intention that the living room of the house would become a sound installation as part of the overall exhibition. Further recordings were made of environmental sounds from each school's locale.

The living room of the house (Figure 7) was small (approximately 3m x 4m) with a passageway marked by the entrance (the front door of the house) and an exit to a hallway as shown in Figure 8. The two sensors used for the installation were placed to make the majority of the room an active sensing space, with the exception of the passageway.

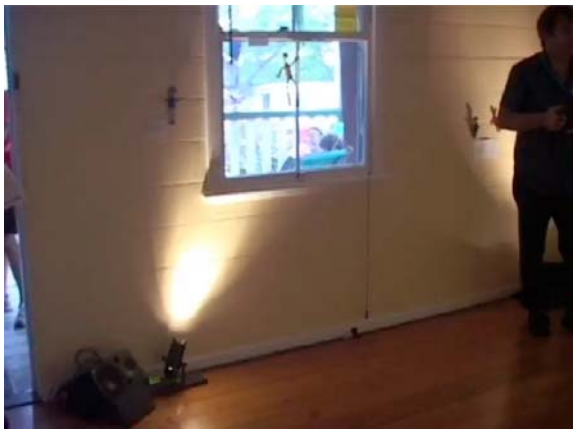


Figure 7 – Come Home Installation Space

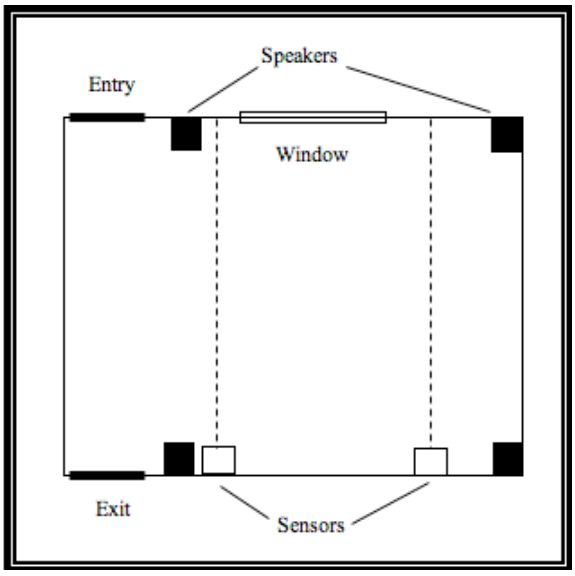


Figure 8 – Come Home Installation Space Floor Plan

The range of the two sensors was divided into two triggering points, providing a total of four triggering points for the entire space, as labelled 1 – 4 in Figure 9.

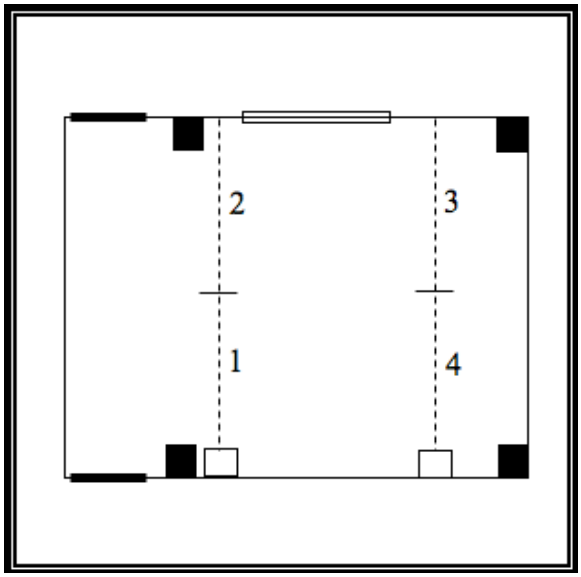


Figure 9 – Come Home Sensor Trigger Points

Even with this minimal number of triggering points, the design of the installation needed to allow for a variety of audio outcomes. The sounds were divided into three categories of environmental sounds, children's' texts and short sound effects, each category triggered independently from the others. This resulted in a constantly changing sound environment, as lengths of the recordings in each category were highly varied.

The environmental recordings contained a wide variety of sounds including birds, traffic, planes and a gramophone playing. Ranging in length from 30" to over 3'00, these formed a sonic background that was triggered at all four triggering points, though limited to one recording being played at any one time. Incoming triggers were also monitored so that should there be a period of any more than 10" without a trigger from the space, any environmental sound still playing would fade out. These sounds were distributed equally to all four speakers in the quad setup.

Text recordings were typically around 30", though could extend to 2'00 in length. These were organised in the same way as the environmental sounds, i.e. triggered at all four triggering points, one text at a time, and faded out when there was no activity in the space. The texts were subject to a simple level of spatialisation wherein the audio would be heard from the nearest speaker to the triggering point. Should a visitor move through another trigger point, the audio would pan to the speaker nearest the triggering point. Using this panning, a text could be triggered and as the visitor moves around the space the text 'follows' the movements.

The sound effects were again triggered at all triggering points, and being short (under 30") did not require any fade out. Similar to the text spatialisation, the sound effects were distributed to the four speakers, however for clarity the sounds were distributed to the speaker opposite the triggering point as opposed to the speaker nearest the triggering point.

The inclusion of this sound installation within the overall exhibition augmented the experience for visitors as they were able to hear the voices of the children and to hear the children speaking about their chosen elders and their relationships with their elders. As the majority of the exhibition was based on the children's presentation of visual art works, the inclusion of this sound installation provided variety and another dimension to the exhibition.

Conclusion

The three installations discussed illustrate the diversity of approaches that may be employed in the design of installations using non-tactile ultrasonic sensing and the variations possible in design according to the installation setting. In Flow, a requirement that the work be suited to its river festival setting resulted in the PLaY+SPaCE system's Activity Mode being utilised to facilitate a programmatic depiction of the course of a river. As a non-tactile system, PLaY+SPaCE allowed the public to move freely throughout the space, using their own levels of activity to realise the programmatic and musical aspects of the work.

In *Riverscape*, the three areas of the exhibition, and the varying art works presented in each, required three different approaches to audio assignment and triggering. The basic premise of Reactivity was used in this work to provide simple audio triggering that could be easily understood and controlled by visitors. The audio materials presented within each area served to heighten visitors' experiences of the art works in each area of the exhibition.

Similarly, the *Come Home* exhibition used simple triggering of audio samples. The varying lengths of the samples used for text, environmental sounds and sound effects ensured that the audio texture created with all three categories of sounds was constantly changing. The multichannel quad setup was beneficial in enhancing the experience of the children's texts through panning, and allowed the distribution of texts and sound effects to different speakers for clarity.

References

Bandt, R. 2005. "Designing Sound in Public Space in Australia: a comparative study based on the Australian Sound Design Project's online gallery and database." In *Organised Sound* 10:2, 129-140.

Bandt, R. 2006. "Sound Installation: Blurring the Boundaries of the Eye, the Ear, Space and Time." In *Contemporary Music Review*, 25:4, 353-365.

Bongers, B. 2000. Physical Interfaces in the Electronic Arts. Interaction Theory and Interfacing Techniques for Real-time Performance. In Wanderly, M. & Battier M. (Eds.) *Trends in Gestural Control of Music.* Paris, IRCAM - Centre Pompidou.

Campbell, S. 2003. "Bats, Max, Boids and Music." *Proceedings of the 2003 Apple University Consortium Conference*. Available from auc.uow.edu.au/conf/conf03/papers/AUC_DV2003_Campbell.pdf [Accessed 18th March, 2005].

Campbell, S. 2005. "PLaY+SPaCE: An Ultrasonic Gestural MIDI Controller." *Proceedings of the 2005 Australasian Computer Music Conference*: Australasian Computer Music Association: 43-49.

Manning, P. 2004. *Electronic and Computer Music*. Oxford: Oxford University Press.

Manovich, L. 2001. *The Language of New Media.* Cambridge, Mass. MIT Press.

Winkler, T. 1998. *Composing Interactive Music, Techniques and Ideas Using Max.* Massachusetts. The MIT Press.